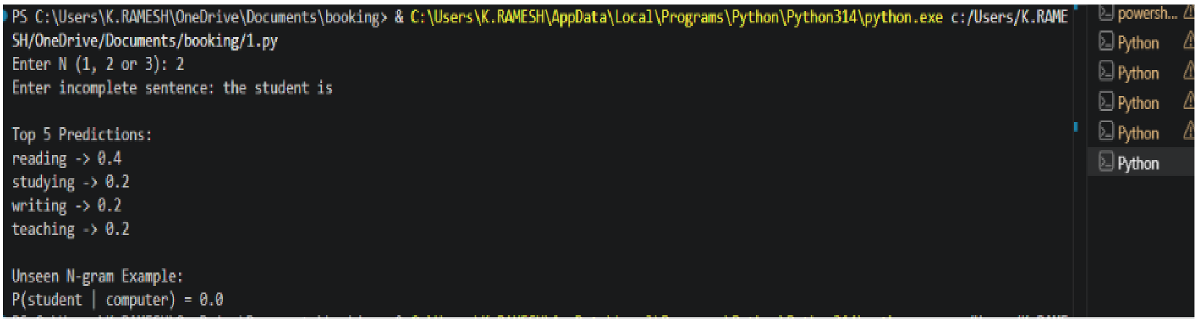
N-gram Language Model

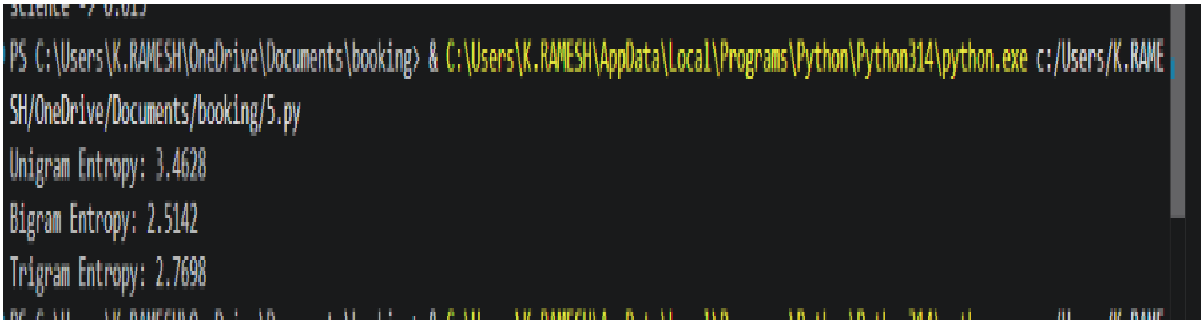


```
PS C:\Users\K.RAMESH\OneDrive\Documents\booking> & C:\Users\K.RAMESH\AppData\Local\Programs\Python\Python314\python.exe c:/Users/K.RAME
SH/OneDrive/Documents/booking/1.py
Enter N (1, 2 or 3): 2
Enter incomplete sentence: the student is

Top 5 Predictions:
reading -> 0.4
studying -> 0.2
writing -> 0.2
teaching -> 0.2

Unseen N-gram Example:
P(student | computer) = 0.0
```

Backoff and Deleted Interpolation

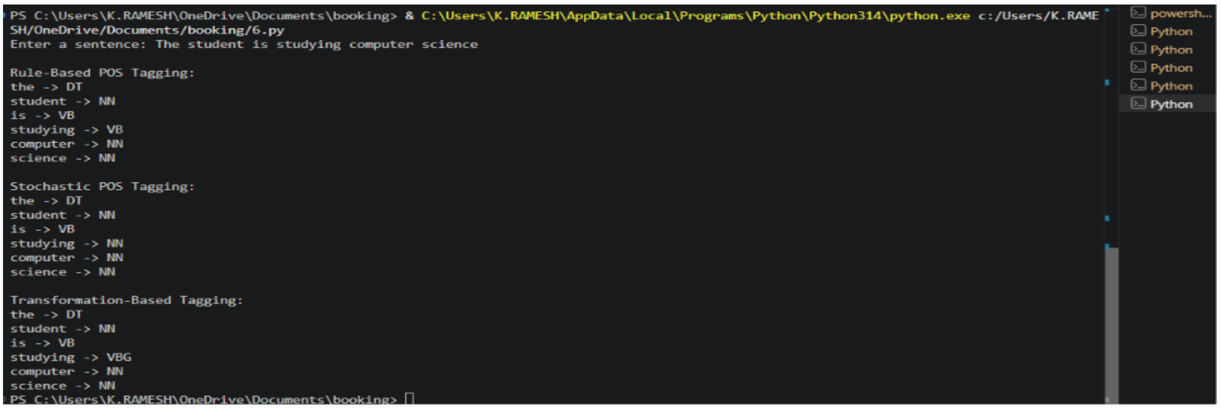
```
PS C:\Users\K.RAMESH\OneDrive\Documents\booking> & C:\Users\K.RAMESH\AppData\Local\Programs\Python\Python314\python.exe c:/Users/K.RAME
SH/OneDrive/Documents/booking/3.py
Enter sentence: The teacher is reading a book

Unsmoothed Predictions:
the -> 0.0
student -> 0.0
is -> 0.0
studying -> 0.0
computer -> 0.0

Backoff Predictions:
the -> 0.175
student -> 0.125
is -> 0.125
computer -> 0.075
science -> 0.075

Deleted Interpolation Predictions:
the -> 0.035
student -> 0.025
is -> 0.025
computer -> 0.015
science -> 0.015
```

Entropy of N-gram Model



```
PS C:\Users\K.RAMESH\OneDrive\Documents\booking> & C:\Users\K.RAMESH\AppData\Local\Programs\Python\Python314\python.exe c:/Users/K.RAME
SH/OneDrive/Documents/booking/5.py
Unigram Entropy: 3.4628
Bigram Entropy: 2.5142
Trigram Entropy: 2.7698
```

POS Tagging



```
PS C:\Users\K.RAMESH\OneDrive\Documents\booking> & C:\Users\K.RAMESH\AppData\Local\Programs\Python\Python314\python.exe c:/Users/K.RAME
SH/OneDrive/Documents/booking/6.py
Enter a sentence: The student is studying computer science

Rule-Based POS Tagging:
the -> DT
student -> NN
is -> VB
studying -> VB
computer -> NN
science -> NN

Stochastic POS Tagging:
the -> DT
student -> NN
is -> VB
studying -> NN
computer -> NN
science -> NN

Transformation-Based Tagging:
the -> DT
student -> NN
is -> VB
studying -> VBG
computer -> NN
science -> NN
PS C:\Users\K.RAMESH\OneDrive\Documents\booking>
```